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Network communication device having electromagnetic shielding function

Abstract

A network communication device has a circuit board, a network communication chip, an antenna, and a metal cover. The circuit board has a first surface and a second surface opposite to each other. An interior of the circuit board has a ground layer. An edge of the first surface has a ground ring. The ground ring surrounds the first surface and is electrically connected to the ground layer. The network communication chip is disposed on the first surface. The antenna is electrically connected to the circuit board. The metal cover has an annular support rib. The circuit board is disposed on the annular support rib with the first surface facing the metal cover. The ground ring is in physical contact with and electrically connected to the annular support rib.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of China application serial no. 202210682091.6, filed on Jun. 16, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a network communication device, in particular to a network communication device having an electromagnetic shielding function.

Description of Related Art

(3) With the advancement of technology, the number of connected components on the circuit board has increased, and the transmission performance has also improved day by day. However, the fast operating performance is likely to generate electromagnetic waves and affect the operation of other electronic components. For example, an antenna connected to a circuit board is susceptible to electromagnetic interference from other electronic components, which may affect signal reception. In the related art, metal shields are mostly used to shield the network communication chip to shield electromagnetic waves, but the shielding effect is still limited. How to design a simple shield to shield electromagnetic waves well and maintain the operating performance of electronic components is a direction that persons skilled in the art are devoting to research and develop.

SUMMARY

(4) The disclosure is a network communication device having an electromagnetic shielding function, which can improve the effectiveness of the antenna for receiving signals.

(5) According to an embodiment of the disclosure, a network communication device includes a circuit board, a network communication chip, an antenna, and a first metal cover. The circuit board has a first surface and a second surface opposite to each other. An interior of the circuit board has a ground layer. An edge of the first surface has a ground ring. The ground ring surrounds the first surface and is electrically connected to the ground layer. The network communication chip is disposed on the first surface. The antenna is electrically connected to the circuit board. The first metal cover has an annular support rib. The circuit board is disposed on the annular support rib with the first surface facing the first metal cover. The ground ring is in physical contact with and electrically connected to the annular support rib.

(6) According to an embodiment of the disclosure, the first metal cover further has a retaining wall, and the annular support rib and a contact surface of the ground ring are shielded by the retaining

wall when viewed along a direction parallel to the first surface.

(7) According to an embodiment of the disclosure, the circuit board further includes a plurality of solder balls. The plurality of solder balls are arranged in a staggered manner on the ground ring and connected to the ground ring.

(8) According to an embodiment of the disclosure, diameters of the plurality of solder balls respectively range from 11 mils to 13 mils.

(9) According to an embodiment of the disclosure, spacings between the plurality of solder balls respectively range from 9 mils to 11 mils.

(10) According to an embodiment of the disclosure, the network communication device further includes a second metal cover assembled with the first metal cover, in which the circuit board is disposed between the first metal cover and the second metal cover, and the antenna is positioned outside the first metal cover and the second metal cover.

(11) According to an embodiment of the disclosure, the first metal cover and the second metal cover are entirely made of metal.

(12) According to an embodiment of the disclosure, the ground layer is electrically connected to the ground ring through a plurality of plated through holes.

(13) According to an embodiment of the disclosure, the network communication device further includes an electromagnetic wave shielding cover disposed on the first surface and covering the network communication chip. The first metal cover further has a heat dissipation portion, and the heat dissipation portion is in thermal contact with the electromagnetic wave shielding cover.

(14) According to an embodiment of the disclosure, the circuit board further includes a connection port, in which a metal shell of the connection port is in physical contact with and electrically connected to the annular support rib.

(15) According to an embodiment of the disclosure, the first surface also has a bare metal layer, the first metal cover further has a heat dissipation portion, the heat dissipation portion is in thermal contact with the bare metal layer, and the bare metal layer is electrically connected to the ground layer.

(16) According to an embodiment of the disclosure, a width of the ground ring is greater than or equal to 1.5 mm.

(17) Based on the above, in the network communication device of the disclosure, the first metal cover and the circuit board can effectively shield electromagnetic waves generated by the network communication chip, so that the antenna can receive external signals well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic view of a network communication device according to an embodiment of the disclosure.

(2) FIG. 1B is a schematic exploded view of the network communication device in FIG. 1A.

(3) FIG. 2A is a bottom view of the circuit board of the network communication device in FIG. 1B.

(4) FIG. 2B is a top view of the first metal cover of the network communication device in FIG. 1B.

(5) FIG. 3 is a partially enlarged cross-sectional view of the circuit board in FIG. 2A.

(6) FIG. 4 is a partially enlarged view of a ground ring of the network communication device in FIG. 2A.

(7) FIG. 5A is a schematic cross-sectional view of the network communication device of FIG. 1A along a section line A-A.

(8) FIG. 5B is a schematic cross-sectional view of another embodiment of the disclosure.

(9) FIG. 5C is a schematic cross-sectional view of another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(10) Reference will now be made in detail to the exemplary embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used in the drawings and description to refer to the same or like parts.

(11) The network communication devices of the various embodiments of the disclosure are, for example, wireless access points using 6 GHz unlicensed frequency bands, indoor millimeter wave wireless communication devices, intelligent 10 Gigabit optical fiber routers based on deep message detection, and high-definition large-screen smart phone conference devices based on Android systems, gigabit mobile access routers supporting mobile app management, but the disclosure is not limited thereto. In addition, the network communication device of the disclosure may be equipped with various forms of appearance parts and other components according to requirements, and the disclosure is not limited thereto.

(12) FIG. 1A is a schematic view of a network communication device according to an embodiment of the disclosure. FIG. 1B is a schematic exploded view of the network communication device in FIG. 1A. FIG. 2A is a bottom view of the circuit board of the network communication device in FIG. 1B. Please refer to FIG. 1A to FIG. 2A. A network communication device **100** of this embodiment includes a circuit board **110**, a network communication chip **120**, and an antenna **130**. The circuit board **110** has a first surface **S1** and a second surface **S2** opposite to each other, and an edge of the first surface **S1** of the circuit board **110** has a ground ring **112b**. An interior of the circuit board **110** has a ground layer **112a**, the ground layer **112a** is positioned between the first surface **S1** and the second surface **S2**, and the ground ring **112b** of the circuit board **110** surrounds the first surface **S1** and is electrically connected to the ground layer **112a**. The antenna **130** is electrically connected to the circuit board **110**. The network communication chip **120** is disposed on the first surface **S1** of the circuit board **110**. The network communication chip **120** is, for example, a chip such as an RF chip, a transceiver chip, a baseband chip, a System on a Chip, and the kind of network communication chip **120** of the disclosure is not limited. For example, the ground ring **112b** surrounds the network communication chip **120**.

(13) FIG. 2B is a top view of the first metal cover of the network communication device in FIG. 1B. Please refer to FIG. 1A, FIG. 1B, and FIG. 2B. The network communication device **100** of this embodiment includes a first metal cover **140**, and the first metal cover **140** has an annular support rib **142**. When the circuit board **110** is assembled on the first metal cover **140**, the circuit board **110** is disposed on the annular support rib **142** with the first surface **S1** facing the first metal cover **140**. The ground ring **112b** is in physical contact with and electrically connected to the annular support rib **142**, that is, a current may flow in between.

(14) According to the above, the ground layer **112a** of the circuit board **110** is electrically connected to the annular support rib **142** of the first metal cover **140** through the ground ring **112b**, and the first metal cover **140** and the circuit board **110** seal the network communication chip **120** in a space formed by the first metal cover **140** and the circuit board **110**. The ground layer **112a** is substantially spread over all or most areas of the circuit board **110**, so any area of the circuit board **110** where electromagnetic waves arrive is received by the ground layer **112a**. In addition, any area of the first metal cover **140** where the electromagnetic waves arrive is also received. The first metal cover **140** is connected to the ground layer **112a** through the ground ring **112b** to form a common ground. Such a design allows the first metal cover **140** and the ground layer **112a** of the circuit board **110** to jointly form a shielding chamber to shield electromagnetic waves generated during an operation process of the network communication chip **120**, so that the electromagnetic waves cannot radiate from the shielding chamber to outside, and external electromagnetic waves cannot enter the shielding chamber. Moreover, the network communication device **100** of this embodiment only needs to dispose electrical connection to the ground ring **112b** of the ground layer **112a** on the circuit board **110**, and the effect of electromagnetic shielding can be achieved with the first metal cover **140**. That is to say, the manufacturing cost of the network communication device **100** of this embodiment is low and the structure is simple. In this embodiment, when assembling the network

communication device **100**, only the ground ring **112b** needs to be in physical contact with the annular support rib **142**, and no clamping structure or adhesive is required between the ground ring **112b** and the annular support rib **142** to connect the ground ring **112b** and the annular support rib **142**. The process is fast and convenient, and the disassembling in the future is also convenient. In other embodiments, the network communication device **100** may also use appropriate fastening structures, locking structures, or adhesives to respectively fix each component of the network communication device **100**.

(15) In the embodiment of the disclosure, the antenna **130** is disposed on the second surface **S2** of the circuit board **110** and exposed outside the space formed by the first metal cover **140** and the circuit board **110**. That is to say, the network communication chip **120** and the antenna **130** are respectively disposed on two opposite surfaces of the circuit board **110**. At this time, since electromagnetic waves of the network communication chip **120** is shielded by the first metal cover **140** and the circuit board **110**, the reception of the antenna **130** is less likely to be influenced by the electromagnetic waves, and external signals can be received well. The first metal cover **140** may be combined with another cover to form a housing of the network communication device **100**. The antenna **130** may be positioned in a space in the housing but outside the shielding chamber, such as on the second surface **S2** or a space between the second surface **S2** and another cover, or may also be an external antenna disposed at an outer side of the housing. Also, the first metal cover **140** may not be positioned at an outer side of the network communication device **100**, but the components shown in FIG. **1B** are covered by the unshown housing.

(16) In addition, other components that are likely to generate electromagnetic waves may also be disposed on the first surface **S1** of the circuit board **110**, so that these components are positioned in the conductive shielding chamber formed by the ground layer **112a** and the first metal cover **140**, so as to shield electromagnetic waves generated during an operation process of the components without radiating to the outside. On the other hand, components that are less likely to generate electromagnetic waves may be disposed on the second surface **S2** of the circuit board **110**.

(17) FIG. **3** is a partially enlarged cross-sectional view of the circuit board in FIG. **2A**. For convenience of illustration, the sizes of the ground layer **112a**, the ground ring **112b**, and a plated through hole **112c** in FIG. **3** are all enlarged, but are not used to represent the actual size. Referring to FIG. **3**, in an embodiment of the disclosure, the ground layer **112a** is electrically connected to the ground ring **112b** through a plurality of plated through holes **112c**. The first metal cover **140** may be entirely made of metal. Through the above design, when the annular support rib **142** of the first metal cover **140** is in physical contact with the ground ring **112b** of the circuit board **110**, the ground layer **112a** and the first metal cover **140** form a conductive shielding chamber, so that the electromagnetic waves of the network communication chip **120** on the circuit board **110** does not affect the antenna **130**, and the external electromagnetic waves does not affect the network communication chip **120**. In other embodiments, the first metal cover **140** may be partially made of non-metal, as long as the metal components may be distributed in most areas and conduct with each other to shield electromagnetic waves, the disclosure is not limited thereto.

(18) FIG. **5A** is a schematic cross-sectional view of the network communication device in FIG. **1A** along a section line A-A. For the convenience of illustration, the size of some components in FIG. **5A** is enlarged, but is not used to represent the actual size or the size proportion with other components. Please refer to FIG. **5A**. In the embodiment of the disclosure, the network communication device **100** further includes an electromagnetic wave shielding cover **160**. The electromagnetic wave shielding cover **160** is disposed on the first surface **S1** and covering the network communication chip **120**, so that electromagnetic waves of the network communication chip **120** is less likely to radiate outside the space formed by the first metal cover **140** and the circuit board **110**, and can also avoid interfering other surrounding electronic components or being interfered by electromagnetic waves from the surrounding electronic components.

(19) In an embodiment of the disclosure, the first metal cover **140** further has a heat dissipation

portion **148a**. The heat dissipation portion **148a** is in thermal contact with the electromagnetic wave shielding cover **160** and the network communication chip **120**. That is to say, the heat dissipated during the operation of the network communication chip **120** may be conducted to the heat dissipation portion **148a** and the entire first metal cover **140**, thereby maintaining the performance of the network communication chip **120** and avoiding damage to the network communication chip **120**. In addition, a heat dissipation pad **170** may be provided between the network communication chip **120** and the electromagnetic wave shielding cover **160** and between the heat dissipation portion **148a** and the electromagnetic wave shielding cover **160** to further improve the efficiency of heat conduction.

(20) In addition, in the embodiment of the disclosure, the first surface **S1** also has a bare metal layer **118**, and the first metal cover **140** further has a heat dissipation portion **148b**. The heat dissipation portion **148b** is in thermal contact with the bare metal layer **118**, and the bare metal layer **118** is electrically connected to the ground layer **112a**. Such a design allows the heat on the circuit board **110** to be thermally conducted to the heat dissipation portion **148b** and the entire first metal cover **140** through the bare metal layer **118**, thereby maintaining the performance of components disposed on the circuit board **110**. In addition, the heat dissipation pad **170** may also be provided between the heat dissipation portion **148b** and the bare metal layer **118** to further improve the efficiency of heat conduction. The bare metal layer **118** may be formed together with the ground ring **112b** and other lines on the circuit board **110**. The bare metal layer **118** may be formed on the first surface **S1** of the circuit board **110** in an area where no other components are disposed, and the position of the bare metal layer **118** may be opposite to positions disposed with heat-prone components on the second surface **S2** of the circuit board **110**, so as to efficiently dissipate heat from the heat-prone components.

(21) It should be noted that the heat dissipation portion **148a** and the heat dissipation portion **148b** in this embodiment are part of the first metal cover **140**. In other embodiments, the heat dissipation portion **148a** and the heat dissipation portion **148b** may also be external components that are in thermal contact with the first metal cover **140**. In addition, the heat dissipation pad **170** may also be replaced by heat dissipation paste, heat conduction glue, or other materials or components that are easy to conduct heat, and the disclosure is not limited thereto.

(22) Please refer to FIG. 1B and FIG. 2A. In the embodiment of the disclosure, the circuit board **110** further includes a connection port **116**. A metal shell of the connection port **116** is in physical contact with and electrically connected to the annular support rib **142**. The connection port **116** is electrically connected to the internal circuit of the circuit board **110** and the ground layer **112a**. Through the above design, even if the ground ring **112b** has an opening at the place where the connection port **116** is disposed on the circuit board **110**, so that the ground ring **112b** does not completely surround the circuit board **110**, the circuit board **110** and the first metal cover **140** may still form a substantially complete shielding chamber.

(23) In addition, the first metal cover **140** also has a through port **146**. When the circuit board **110** is assembled on the first metal cover **140**, the position of the connection port **116** corresponds to the through port **146** of the first metal cover **140**, so that the user can still electrically connect external components to the circuit board **110** through the through port **146**.

(24) FIG. 4 is a partially enlarged view of a ground ring of the network communication device in FIG. 2A. Please refer to FIG. 4. In the embodiment of the disclosure, there are also a plurality of solder balls **114** arranged in a staggered manner on the ground ring **112b** and connected to the ground ring **112b**. In other words, the ground ring **112b** of the circuit board **110** is in physical contact with and electrically connected to the annular support rib **142** of the first metal cover **140** through the plurality of solder balls **114**. It should be noted that, for the clarity of the drawing, the plurality of solder balls **114** are not shown on the ground ring **112b** in FIG. 2A.

(25) The plurality of solder balls **114** are respectively arranged along a first direction **F1** and a second direction **F2**. The first direction **F1** is, for example, the extending direction of the ground

ring **112b**. The second direction **F2** is perpendicular to the first direction **F1**, and the second direction **F2** is, for example, parallel to the direction in which the network communication chip **120** emits electromagnetic waves outward. As shown in FIG. 4, looking along the second direction **F2** parallel to the surface of the circuit board **110**, another solder ball **114** is seen between the solder balls **114**. In other words, when the network communication chip **120** emits electromagnetic waves outward, the electromagnetic waves touches at least one solder ball **114** and is received. Through the solder balls **114** arranged in a staggered manner on the ground ring **112b** as described above, the electromagnetic waves emitted by the network communication chip **120** does not directly radiate outward from the gap between the ground ring **112b** and the first metal cover **140** to affect the reception of the antenna **130**. Experiments have proved that by designing diameters **D** of the solder balls **114** to respectively range from 11 mils to 13 mils, such as 12 mils, spacings **L** between the solder balls **114** to respectively range from 9 mils to 11 mils, such as 10 mils, and a width **W** of the ground ring **112b** to be greater than or equal to 1.5 mm, electromagnetic waves emitted from the network communication chip **120** to outside the first metal cover **140** and the circuit board **110** can be reduced, and the reception of the antenna **130** is not affected.

(26) Since the solder balls **114** are less likely to be oxidized than the ground ring **112b**, after the solder balls **114** are melted on the ground ring **112b** through reflow, the ground ring **112b** may be stably electrically connected to the first metal cover **140** through the less likely to be oxidized solder balls **114**. In addition, the solder ball **114** is not fixedly bonded to the annular support rib **142** of the first metal cover **140**, and there is no interlocking structure between the annular support rib **142** and the circuit board **110**, so the circuit board **110** can be easily assembled or disassembled from the first metal cover **140** repeatedly.

(27) It should be noted that, although FIG. 4 only shows a part of the ground ring **112b** and the solder balls **114** disposed thereon, the entire ground ring **112b** is disposed with the solder balls **114** as shown in FIG. 4. In addition, the arrangement of the solder balls **114** is not limited to the arrangement shown in FIG. 4. For example, the solder balls **114** may be arranged in four or more rows parallel to the first direction **F1**. The shape of the solder ball **114** may be round, square, rhombus, or other shapes.

(28) Please refer to FIG. 5A. The first metal cover **140** of this embodiment further has a retaining wall **144**. The annular support rib **142** and a contact surface of the ground ring **112b** are shielded by the retaining wall **144** when viewed along a direction **F3** parallel to the first surface **S1**. As shown in FIG. 1A and FIG. 5A, when the circuit board **110** is assembled on the first metal cover **140**, the retaining wall **144** completely surrounds the circuit board **110**. Such a design makes it less likely for the electromagnetic waves emitted by the network communication chip **120** to enter or exit the shielding chamber jointly formed by the first metal cover **140** and the circuit board **110** to radiate from between the circuit board **110** and the first metal cover **140** to the outside.

(29) In this embodiment, the retaining wall **144** is shown integrally disposed on the outer periphery of the first metal cover **140**. In other embodiments of the disclosure, the retaining wall **144** may also be disposed inside the outer periphery of the first metal cover **140** according to the size of the circuit board **110**, and the retaining wall **144** still completely surrounds the circuit board **110**. Such a design makes it less likely for the electromagnetic waves emitted by the network communication chip **120** to enter or exit the shielding chamber jointly formed by the first metal cover **140** and the circuit board **110** to radiate from between the circuit board **110** and the first metal cover **140** to the outside.

(30) FIG. 5B is a schematic cross-sectional view of another embodiment of the disclosure. Please refer to FIG. 5A and FIG. 5B. A network communication device **100a** of this embodiment is substantially the same as the network communication device **100** of FIG. 5A. The difference between the two is that, the network communication device **100a** of this embodiment further includes a second metal cover **150**. The second metal cover **150** is assembled with the first metal cover **140**, and the circuit board **110** is disposed between the first metal cover **140** and the second

metal cover **150**. Since the second metal cover **150** is also entirely made of metal, the shielding of the electromagnetic waves generated during the operation of the network communication chip **120** may be further strengthened. At this time, an antenna **130a** is positioned outside the first metal cover **140** and the second metal cover **150**.

(31) FIG. 5C is a schematic cross-sectional view of another embodiment of the disclosure. Please refer to FIG. 5A and FIG. 5C. A network communication device **100b** of this embodiment is substantially the same as the network communication device **100** shown in FIG. 5A. The difference between the two is that, the network communication device **100b** of this embodiment does not adopt the electromagnetic wave shielding cover **160** to cover the network communication chip **120**, but a heat dissipation portion **148c** is in thermal contact with the network communication chip **120** through the heat dissipation pad **170**, so as to improve the efficiency of heat conduction. In another embodiment of the disclosure, the network communication device **100** does not adopt the electromagnetic wave shielding cover **160** to cover the network communication chip **120**, but the heat dissipation portion **148c** is directly in thermal contact with the network communication chip **120**, so as to improve the efficiency of heat conduction.

(32) Based on the above, in the network communication device of the disclosure, the annular support rib of the first metal cover is electrically connected to the ground ring of the circuit board, so that the first metal cover and the circuit board can shield electromagnetic waves, so that the electromagnetic waves are less likely to enter or exit the chamber formed by the first metal cover and the circuit board. The retaining wall of the first metal cover and the solder ball on the ground ring can further shield electromagnetic waves. In addition, the circuit board is not fixedly bonded to the first metal cover, so that users can easily assemble or disassemble the circuit board from the first metal cover repeatedly.

(33) Finally, it should be noted that the above embodiments are only used to illustrate the technical solutions of the disclosure, rather than to limit them. Although the disclosure has been described in detail with reference to the foregoing embodiments, persons skilled in the art should understand that the technical solutions described in the foregoing embodiments may still be modified, or some or all of the technical features may be equivalently replaced. However, the modifications or replacements do not make the essence of the corresponding technical solutions depart from the scope of the technical solutions of the embodiments of the disclosure.

Claims

1. A network communication device, comprising: a circuit board having a first surface and a second surface opposite to each other, wherein an interior of the circuit board has a ground layer, an edge of the first surface has a ground ring, and the ground ring surrounds the first surface and is electrically connected to the ground layer; a network communication chip disposed on the first surface; an antenna electrically connected to the circuit board; a first metal cover having an annular support rib, wherein the circuit board is disposed on the annular support rib with the first surface facing the first metal cover, and the ground ring is in physical contact with and electrically connected to the annular support rib; and an electromagnetic wave shielding cover disposed on the first surface and covering the network communication chip, the first metal cover further has a heat dissipation portion in thermal contact with the electromagnetic wave shielding cover.
2. The network communication device according to claim 1, wherein the first metal cover further has a retaining wall, and the annular support rib and a contact surface of the ground ring are shielded by the retaining wall when viewed along a direction parallel to the first surface.
3. The network communication device according to claim 1, further comprising a plurality of solder balls arranged in a staggered manner on the ground ring and connected to the ground ring.
4. The network communication device according to claim 3, wherein diameters of the plurality of solder balls respectively range from 11 mils to 13 mils, and spacings between the plurality of solder

balls respectively range from 9 mils to 11 mils.

5. The network communication device according to claim 1, wherein the first metal cover further has a heat dissipation portion in thermal contact with the network communication chip.

6. The network communication device according to claim 1, further comprising a second metal cover assembled with the first metal cover, wherein the circuit board is disposed between the first metal cover and the second metal cover, and the antenna is positioned outside the first metal cover and the second metal cover.

7. The network communication device according to claim 6, wherein the first metal cover and the second metal cover are entirely made of metal.

8. The network communication device according to claim 1, wherein the ground layer is electrically connected to the ground ring through a plurality of plated through holes.

9. The network communication device according to claim 1, wherein the circuit board further comprises a connection port, and a metal shell of the connection port is in physical contact with and electrically connected to the annular support rib.

10. The network communication device according to claim 1, wherein the first surface also has a bare metal layer, the heat dissipation portion is in thermal contact with the bare metal layer, and the bare metal layer is electrically connected to the ground layer.

11. The network communication device according to claim 1, wherein a width of the ground ring is greater than or equal to 1.5 mm.
